

# A Proof of Written on the Wall II Conjecture 146

A graph-metric inequality certified in Lean 4

**Theorem.** Let  $G$  be a finite, nontrivial, connected simple graph. Let  $B$  be its set of peripheral vertices; let  $p$  be the largest distance from a vertex to  $B$ ; let  $t$  be the order of a largest induced tree; and let  $\rho$  be the radius of the square graph  $G^2$ . Then

$$2p \leq t \rho$$

## Notation and elementary facts

Write  $r = \text{rad}(G)$  and  $d = \text{diam}(G)$ . The square graph  $G^2$  has the same vertices as  $G$ , with two distinct vertices adjacent exactly when their  $G$ -distance is at most two. A geodesic is a shortest path. Every geodesic is induced: a chord between two nonconsecutive vertices would shorten it.

**Lemma 1 (the square metric).** For all vertices  $a, b$ ,  $\text{dist}_{G^2}(a, b) = \text{ceil}(\text{dist}_G(a, b)/2)$ ; consequently  $\rho = \text{ceil}(r/2)$ .

Each edge of  $G^2$  spans at most two  $G$ -edges, proving the lower bound. Conversely, traverse a  $G$ -geodesic two edges at a time to obtain a  $G^2$ -walk of the stated length. Taking the maximum over  $b$  and then the minimum over  $a$  yields the radius identity.

**Lemma 2 (global bounds).**  $r \leq d \leq 2r$ ,  $t \geq d+1$ , and  $p \leq d-1$ .

The radius-diameter bounds are standard. A diametral geodesic is an induced path on  $d+1$  vertices, hence gives  $t \geq d+1$ . For the final inequality,  $p \leq d$  because  $B$  is nonempty. If  $d(z, B) = d$ , then every peripheral vertex is distance  $d$  from  $z$ , so  $\text{ecc}(z) = d$  and  $z$  is peripheral. Thus  $z$  lies in  $B$  and  $d(z, B) = 0$ , contradicting  $d(z, B) = d > 0$ .

## Reduction to one exceptional configuration

By Lemma 1,  $\rho = \text{ceil}(r/2)$ . If  $r \geq 3$ , then  $\rho \geq 2$  and

$$t \rho \geq 2(d+1) \geq 2(d-1) \geq 2p.$$

If  $r=1$ , then  $d \leq 2$ ,  $p \leq 1$ ,  $\rho=1$ , and  $t \geq 2$ . Now let  $r=2$ . Then  $\rho=1$  and  $d \leq 4$ . For  $d \leq 3$ , the inequalities  $2p \leq 2(d-1) \leq d+1 \leq t$  suffice. For  $d=4$  and  $p \leq 2$ , one has  $2p \leq 4 \leq 5 \leq t$ . Thus only  $r=2$ ,  $d=4$ ,  $p=3$  remains.

**Exceptional lemma.** If  $r=2$ ,  $d=4$ , and  $p=3$ , then  $G$  contains an induced tree on at least six vertices.

## Proof of the exceptional lemma

Choose  $z$  with  $d(z, B) = 3$ . Then  $\text{ecc}(z) = 3$ : it is at least three,  $z$  is not peripheral, and the diameter is four. Therefore  $d(z, q) = 3$  for every  $q$  in  $B$ . Choose a diametral pair  $x, y$  and a center  $c$ . Then  $d(x, y) = 4$ ,  $d(z, x) = d(z, y) = 3$ ,  $d(c, x) = d(c, y) = 2$ , and  $d(c, z)$  is either one or two.

**Case 1:  $d(c,z)=1$ .** Choose midpoints  $u,v$  with  $x-u-c-v-y$  a geodesic. The vertex  $z$  is adjacent to  $c$  and cannot be adjacent to  $x,y,u$ , or  $v$ : each such edge would put  $z$  within two steps of  $x$  or  $y$ . Hence  $\{x,u,c,v,y,z\}$  induces the six-vertex tree with edges  $xu, uc, cv, vy, cz$ .

**Case 2:  $d(c,z)=2$ .** Choose midpoints  $u,v,w$  with  $x-u-c$ ,  $y-v-c$ , and  $z-w-c$  geodesics. The prescribed distances force every edge among  $\{x,u,c,v,y,w,z\}$  except the six arm edges to be absent, with only  $uw$  and  $vw$  possibly remaining. Indeed,  $u-v$  would shorten the  $x-y$  distance; and triangle inequalities give  $d(u,z), d(v,z), d(w,x), d(w,y) \geq 2$ .

If neither  $uw$  nor  $vw$  is present, all seven vertices induce a tree. If both are present, delete  $c$ ; the six vertices  $\{x,u,w,z,v,y\}$  induce the tree  $xu, uw, wz, wv, vy$ .

## The final chord case

It remains that exactly one of  $uw,vw$  is present; by symmetry take  $uw$ . Then  $x-u-w-z$  is a geodesic. Let  $z-b-a-y$  be a  $z-y$  geodesic.

If  $b=w$ , then  $\{x,u,w,z,a,y\}$  induces the tree  $xu, uw, wz, wa, ay$ . Suppose  $b$  is different from  $w$ . The walk

$$x - u - w - z - b - a - y$$

has seven distinct vertices. Geodesicity and the three terminal distances  $d(x,z)=d(z,y)=3$  and  $d(x,y)=4$  rule out every chord except possibly  $ub$ ,  $wa$ , and  $wb$ . This is the only distance bookkeeping needed in the proof.

| Possible edge                   | Induced tree obtained                               |
|---------------------------------|-----------------------------------------------------|
| $ub$                            | $\{x,u,b,z,a,y\}$ , with edges $xu, ub, bz, ba, ay$ |
| $wa$ (and not $ub$ )            | $\{x,u,w,z,a,y\}$ , with edges $xu, uw, wz, wa, ay$ |
| $wb$ (and neither earlier edge) | $\{x,u,w,b,a,y\}$ , the induced path $x-u-w-b-a-y$  |
| none                            | the full induced seven-vertex path $x-u-w-z-b-a-y$  |

Every branch yields an induced tree on at least six vertices. This proves the exceptional lemma and completes the theorem.

## Formal verification and provenance

The theorem was formalized against the pinned Lean 4 / Formal Conjectures dependencies. The upstream proof replaces the original statement's placeholder with the exact same theorem body, while preserving the formal statement. The kernel audit for the final theorem uses only the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`; it introduces no `sorry`, `admit`, `native_decide`, or project-specific axiom.

The submitted regression module contains one explicit non-vacuity witness: the six-vertex spider formed by the path  $0-1-2-3-4$  together with a pendant edge  $2-5$ . Lean checks that this graph is connected, has  $r=2$  and  $d=4$ , has peripheral set  $\{0,4\}$ , has  $p=3$ , and satisfies the exceptional theorem's conclusion. No exhaustive finite-graph enumeration is claimed in this document.